

VERRELL'S LAW

A Framework for Non-Markovian Information Paths

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Technical edition	Information Physics and Open Quantum Systems Framework
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Status	Proposed theoretical model. Not established physics and not experimentally confirmed.
Continuity	Preserves the law's central proposition: memory and weighted information can influence what follows.

CORE POSTULATE

Physical evolution need not be sequence-independent. When prior interactions leave persistent, physically instantiated information, sufficiently weighted transitions may alter the memory kernel governing later dynamics and thereby bias subsequent transition probabilities.

History becomes active where retained information changes the accessible paths of a physical system.

SCOPE

The paper develops a falsifiable physics-facing formulation using nonequilibrium thermodynamics, generalized master equations, memory kernels and quantum sensing. It does not assume a universal electromagnetic carrier or a proven quantum mechanism.

ABSTRACT

Verrell's Law proposes that retained information from prior interactions can exert disproportionate, directional influence over later physical evolution. This edition expresses that proposition in the language of information thermodynamics and non-Markovian open-system dynamics. A weighted transition is defined operationally through measurable physical quantities - dissipated work, entropy production, order-parameter displacement, generated correlations and relaxation time - rather than through subjective importance alone.

The formal hypothesis is that, after conventional system and bath variables are controlled, the effective memory kernel of a reduced system contains a reproducible history-weighted contribution. That contribution should predict later transition statistics, survive held-out trials, respond to intervention on the retained state and disappear when the relevant history is erased or decorrelated.

This proposal is not the statement that non-Markovian dynamics exists; that is established. Nor does it infer a new force from Landauer's principle. Information must be physically instantiated, and logically irreversible erasure has thermodynamic cost under defined conditions. The Verrell-specific question is whether a declared weighting variable identifies an additional, reproducible component of physical memory beyond standard material, environmental and instrumental explanations.

Candidate tests use engineered reservoirs, qubits or resonators, NV-diamond magnetometry, nanoSQUID scanning and microcalorimetric erasure protocols. Electromagnetic coupling is treated as a measurable candidate channel, not a default conclusion. Spatial range, carrier, coupling strength and whether any quantum-level extension is needed remain open.

SCIENTIFIC POSITION

Established physics supplies the language and instruments. Verrell's Law supplies a narrower hypothesis about disproportionate informational weighting. Only a discriminating experiment can connect the two.

01 - THE CORE POSTULATE

VERRELL'S LAW

Physical systems do not always evolve through sequence-independent transitions. Where prior interactions leave persistent information, sufficiently weighted transitions may modify the effective memory through which historical states influence later probability distributions.

The proposed chain is:

Stage	Object	Operational meaning
1	Historical state $H(t)$	A prepared system-bath history with measurable correlations or retained structure.
2	Weighted transition $W(t)$	A predeclared physical transition scored by thermodynamic, structural and informational observables.
3	Retention in system or environment	Persistent state, correlation, hysteresis, bath excitation or field configuration.
4	History-dependent kernel	Past states contribute to present reduced dynamics over a non-zero memory timescale.
5	Biased transition distribution	Later transition probabilities differ between matched present reduced states with different controlled histories.

INFORMATION IS PHYSICAL - BUT NOT A FREE-STANDING SUBSTANCE

Landauer's principle and modern information thermodynamics show that information processing is constrained by the physical states that encode it. This supports treating retained information as causally relevant where it is embodied in degrees of freedom, correlations or control states. It does not establish information as a separate material substance, nor permit information to act without a carrier.

LOCALITY AND RANGE

The baseline formulation is local open-system physics: a system exchanges energy and information with an environment. Longer-range electromagnetic coupling may be tested if a source, field component, propagation law and detector response are specified. Quantum nonlocal correlations do not by themselves transmit controllable information or justify macroscopic nonlocal event clustering.

02 - DEFINING A WEIGHTED PHYSICAL TRANSITION

A weighted moment is not defined merely as a high-energy event. Large energy, large entropy and structural importance are different quantities. A low-energy transition can carry a great deal of information; a high-energy event can be informationally unstructured. The experiment must therefore define weight before outcomes are analysed.

Let the transition descriptor be:

WEIGHT VECTOR		
$x_w(t) = [W_{diss}, \text{Sigma}, \text{Delta phi} , \text{Delta } I_{S:E}, \tau_c, R]$		
Symbol	Quantity	Interpretation
W_{diss}	Dissipated work	Energy irreversibly transferred beyond equilibrium free-energy change.
Sigma	Entropy production	Total or estimated irreversibility over the declared transition window.
Delta phi	Order-parameter change	Structural displacement, switching amplitude or phase-transition coordinate.
Delta $I_{S:E}$	Generated correlation	Change in system-environment mutual information or another declared correlation measure.
τ_c	Correlation time	Timescale over which the induced state or correlation remains predictive.
R	Recurrence / resonance	Predeclared spectral, phase or repeated-transition structure.

A scalar index $w(t)$ may be constructed from standardised components using fixed coefficients a :

DECLARED WEIGHT
$w(t) = a^T z[x_w(t)]$

The coefficients, signs, time window and normalisation must be fixed before the confirmatory data are inspected. Alternatively, the components can remain separate to avoid hiding distinct mechanisms inside one score.

KEY BOUNDARY
<i>A transition is weighted because measurable physical features were declared in advance - not because an unusual later result made the transition look important.</i>

03 - NON-MARKOVIAN FORMULATION

For a reduced quantum system with density operator $\rho_S(t)$, a Nakajima-Zwanzig generalized master equation can be written schematically as:

GENERALIZED MASTER EQUATION

$$d\rho_S(t)/dt = L_S(t)\rho_S(t) + I(t) + \int_0^t K(t,\tau)\rho_S(\tau)d\tau$$

L_S is the time-local system contribution; K is the memory superoperator induced by eliminated environmental degrees of freedom; and $I(t)$ represents an inhomogeneous term that can arise from initial system-environment correlations. The memory kernel is not a mysterious added field. In standard theory it is a compact representation of unresolved bath influence on reduced dynamics.

VERRELL-SPECIFIC EXTENSION

The testable extension decomposes the effective kernel into a conventional model and a history-weighted candidate term:

KERNEL DECOMPOSITION

$$K_{eff}(t,\tau | H,W) = K_0(t,\tau | C) + g(w(\tau), H(\tau); \theta) K_V(t,\tau)$$

K_0 is the best supported conventional kernel conditional on measured controls C . K_V is a declared candidate structure. The function g sets how the prepared history and physical weight modulate that structure. Parameters θ must be learned on training data or specified before the confirmatory test.

NULL HYPOTHESIS

NULL

$H_0: g K_V = 0$ within the predeclared equivalence region once standard system, bath, control and instrument memory are modelled.

A non-zero fitted memory kernel alone does not support Verrell's Law. The Verrell-specific contribution must improve out-of-sample prediction, follow the predeclared weighting relation and respond correctly to intervention on H or W .

04 - WHAT WOULD COUNT AS INFORMATION BACKFLOW?

Non-Markovianity has several inequivalent definitions. The paper therefore does not use the term as a vague synonym for memory. Each study must declare its witness and the physical assumptions under which it is valid.

Witness	Signal	Guardrail
Memory-kernel support	A finite fitted $K(t,\tau)$ improves prediction over a time-local model.	Kernel inference can absorb model misspecification or hidden instrument state.
Trace-distance revival	Distinguishability of selected initial states increases during part of the evolution.	Requires controlled state preparation and is witness-specific.
CP indivisibility	Intermediate dynamical maps fail complete-positive divisibility.	Not identical to every other definition of non-Markovianity.
Process tensor	Multi-time interventions reconstruct operational temporal correlations.	Resource-intensive but directly suited to controlled history tests.
Mutual-information revival	System-ancilla or system-environment correlations increase after prior decay.	Correlation growth must be separated from common driving and leakage.

A strong experiment will use at least two compatible views: one model-based measure, such as the kernel or process tensor, and one direct held-out prediction of transition statistics. History A and history B should be prepared so that the reduced system state at comparison time is matched within a stated tolerance while bath or correlation history differs in a controlled way.

IDENTIFIABILITY

- Fix the candidate kernel basis, scale and sign before confirmatory fitting.
- Measure or randomise instrument drift, coloured noise, heating, control-pulse leakage and environmental relaxation.
- Use sham history pulses with matched energy but different informational structure.
- Hold out entire runs, days or devices rather than random points from an autocorrelated trace.
- Require the inferred contribution to reverse or vanish under the registered history intervention.

05 - INFORMATION ERASURE AND THERMODYNAMIC COST

For a symmetric one-bit memory erased quasistatically while coupled to an equilibrium reservoir at temperature T , Landauer's principle gives the familiar lower bound:

LANDAUER BOUND

$$\langle Q \rangle \geq k_B T \ln 2$$

This is a lower bound for a defined logically irreversible operation under defined physical assumptions. Finite-time driving, nonequilibrium initial states, errors, feedback, correlations and reservoir structure can move the measured cost away from the simple expression. Experiments have verified the standard bound in colloidal memories and single-atom quantum systems. A 2026 DRAM experiment showed that inability to prepare thermal equilibrium imposed a tighter practical constraint, while recent nonequilibrium feedback work demonstrates why the equilibrium form must not be applied blindly.

VERRELL-SPECIFIC THERMAL PREDICTION

Detecting $k_B T \ln 2$ during ordinary reset confirms information thermodynamics, not Verrell's Law. A distinctive test compares matched encodings with different controlled histories and asks whether history weight predicts an additional, reproducible term in heat or work after storage reliability, logical entropy, protocol duration and bath state are controlled:

HISTORY-CONDITIONED DISSIPATION

$$\langle Q | H, W \rangle = Q_{base}(protocol, error, bath) + \Delta Q_V(H, W)$$

Support would require ΔQ_V to follow a predeclared direction, survive held-out prediction and change under erasure or decorrelation of the retained history. If conventional nonequilibrium thermodynamics fully accounts for the result, the stronger Verrell term is unnecessary.

STRUCTURAL HYSTERESIS

Magnetic, mechanical and electronic bistable systems offer practical memory substrates. Hysteresis already proves that physical response can depend on history. The Verrell question is narrower: does a declared measure of transition weight explain reproducible residual structure beyond the established hysteresis model?

06 - THE TECHNOLOGY GAP

The technology gap cannot be defined as 'an effect exists below current sensitivity.' It must specify an observable, expected spectral and spatial signature, integration time, noise model and minimum detectable effect. Different quantum sensors trade field sensitivity against spatial resolution and bandwidth.

Instrument	Best role	Reality check
Single / scanning NV centre	Nanoscale magnetic, electric, strain and temperature mapping at ambient conditions.	High spatial resolution; magnetic sensitivity commonly in the nT to microT per square-root-Hz range depending on geometry and readout.
NV ensemble	Bulk or wide-field magnetometry with many centres.	Better field sensitivity, potentially pT-scale in optimised systems, at reduced spatial resolution.
Diamond micro-resonator	Chip-scale enhanced photon collection.	A 2025 demonstration reported 1.0 microT/sqrt(Hz) on-chip and projected 1.3 nT/sqrt(Hz) with waveguide extraction.
nanoSQUID / SQUID-on-tip	Cryogenic local magnetic-moment and flux imaging.	Single-electron-spin-scale moment sensitivity has been demonstrated; field sensitivity and spatial resolution depend on loop size and standoff.
Microcalorimetry / charge counting	Heat, work and single-electron switching in memory operations.	Directly tests thermodynamic accounting rather than inferring it from magnetic fields.

NO UNIVERSAL SUB-PICOTESLA REQUIREMENT

A blanket sub-picotesla threshold would be scientifically empty without source geometry and bandwidth. A nanoscale source can generate a stronger local field over a tiny volume while remaining invisible to a distant bulk sensor. Conversely, excellent bulk sensitivity can average away a local signal. The protocol must calculate the transfer function from candidate source to detector.

DETECTION RULE

Choose the sensor after the candidate carrier predicts what should be measured - not before.

07 - PROTOCOL A: ENGINEERED OPEN QUANTUM SYSTEM

A superconducting qubit, trapped ion, photonic degree of freedom or other controllable open system is coupled to an engineered reservoir with adjustable spectral density and correlation time.

Stage	Requirement
1. Baseline	Characterise time-local dynamics and the standard memory kernel across reservoir settings.
2. History preparation	Prepare bath histories H_A and H_B with matched total injected energy but different temporal or informational structure.
3. State matching	At comparison time t_0 , verify that the reduced system state, temperature and declared observables are equivalent within tolerance.
4. Weighted transition	Apply a registered pulse or structural switch with independently measured $w(t)$. Include energy-matched sham transitions.
5. Readout	Collect multi-time transition statistics; reconstruct a process tensor or fit the registered kernel family.
6. Intervention	Erase, scramble or decorrelate the prepared bath history while preserving present system controls as closely as possible.
7. Replication	Repeat on a second device and with a blinded analysis pipeline.

PRIMARY CONTRAST

TEST STATISTIC

$$\Delta_H = D[P(Y | \rho_S(t_0), H_A, W) || P(Y | \rho_S(t_0), H_B, W)]$$

D may be a preregistered divergence, log-likelihood improvement or outcome contrast. The study supports only ordinary non-Markovianity if H_A and H_B differ but the effect is fully predicted by K_0 . Verrell-specific support requires the additional weighting relation and successful intervention prediction.

FALSIFIER

With adequate power and verified preparation, the registered Verrell term fails if gK_V remains inside the equivalence region, does not generalise, or survives equally under history scrambling and sham conditions.

08 - PROTOCOL B: FIELD AND ERASURE TRACKING

This protocol tests whether a known physical memory substrate produces a local field and thermal signature whose persistence depends on the declared transition weight.

Element	Design
Substrate	A bistable magnetic nanostructure, nanomechanical resonator, quantum dot or electronic memory cell with calibrated state reliability.
Preparation	Create histories with matched final logical state but different weighted transition sequences and matched injected energy where possible.
Field channel	Map magnetic or strain relaxation with an NV probe or nanoSQUID using a registered spatial, temporal and spectral region of interest.
Thermal channel	Measure work and heat during controlled reset with microcalorimetry, charge counting or validated stochastic-thermodynamic inference.
Ablation	Randomise phase, erase intermediate memory, detune a proposed resonance or alter shielding while retaining sham controls.
Outcome	Test whether field persistence, kernel amplitude and erasure cost share the preregistered dependence on physical weight.

PREDICTED PATTERN

- A reproducible relaxation or correlation timescale that differs by prepared history after the final logical state is matched.
- A field signature localised to the substrate or predicted coupling path, not a room-wide unexplained anomaly.
- A weight-response curve fitted on training data and confirmed without retuning.
- Loss or predictable reduction of the effect after erasure, decorrelation, detuning or shielding intervention.
- Thermal accounting consistent with the full nonequilibrium protocol, including errors and feedback.

WHAT WOULD NOT COUNT

An isolated field spike, an uncorrected drift, ordinary hysteresis, heat produced by a reset, or a correlation selected after inspecting many channels would not discriminate Verrell's Law from established physics.

09 - INTEGRATED EVIDENCE LADDER

Level	Evidence	Requirement
Level 1	Formal identifiability	The weighting variable, kernel family, null region and analysis are defined so the model can fail.
Level 2	Controlled history divergence	Matched present reduced states with different randomised histories produce a held-out, directional difference.
Level 3	Intervention and ablation	Erasing, scrambling or altering the retained state changes the effect as predicted.
Level 4	Metrological and thermal convergence	Independent field, state and heat measurements support the same coupling timescale and direction.
Level 5	Independent replication	A separate team reproduces the preregistered effect on another device and rules out the same conventional alternatives.

The ladder is cumulative. A sophisticated equation is not Level 1 evidence unless its parameters can be identified. Sensor detection without causal intervention does not reach Level 3. Landauer-consistent heat is ordinary physics unless it carries the predicted history dependence. Replication must reproduce the effect size and intervention pattern, not merely another instance of non-Markovian dynamics.

DECISION LABELS

Outcome	Meaning
Not supported	The preregistered Verrell term fails its equivalence, direction or replication criteria.
Compatible	A result is consistent with the framework but equally explained by established models.
Provisional support	Levels 1-3 succeed with strong controls and held-out prediction.
Independent support	Level 5 is reached with metrological and analytical independence.

RULE

Established non-Markovianity is the baseline to beat, not evidence to count twice.

10 - CLAIM BOUNDARIES

THE PAPER PROPOSES

- Retained physical information may modify later dynamics through a measurable history-dependent contribution.
- Some transitions may carry disproportionate influence, operationalised through declared thermodynamic, structural and informational quantities.
- A candidate Verrell contribution can be embedded in a generalized master equation and tested against a conventional memory kernel.
- Electromagnetic, resonant and thermal signatures are measurable candidate channels when a source-detector model is supplied.

THE PAPER DOES NOT CLAIM

- That information is an independently established material substance or new fundamental force.
- That non-Markovian dynamics, hysteresis or Landauer heat were discovered by Verrell's Law.
- That every high-energy event is highly weighted or every weighted transition has high entropy.
- That a Mori-Zwanzig kernel proves non-local memory, consciousness physics or modified quantum mechanics.
- That NV centres generically achieve sub-picotesla nanoscale sensitivity or detect single-photon magnetic fields.
- That heat at the Landauer bound proves a Verrell-specific historical trajectory.
- That the carrier is necessarily electromagnetic, quantum or spatially non-local.

CURRENT SCIENTIFIC POSITION

STATUS

Verrell's Law is a proposed theoretical model. Its physics-facing formulation is falsifiable, but no independent experiment has yet isolated the declared history-weighted kernel contribution.

A negative result in a well-defined regime is informative. It constrains the coupling magnitude, memory timescale, observable or carrier in that regime. A positive result must first survive standard open-system, nonequilibrium, control and detector explanations before it is attributed to the law.

11 - CONCLUSION

The heart of Verrell's Law remains unchanged: memory and weighted information can influence what follows. This technical edition asks a direct laboratory question. When two reduced systems appear the same now but their controlled physical histories differ, does a predeclared measure of informational weight identify a reproducible difference in the memory kernel governing their future evolution?

Open quantum systems provide the correct formal arena because they already contain system-environment coupling, memory, initial correlations and information backflow. That foundation is powerful precisely because it prevents ordinary non-Markovianity from being mistaken for new evidence. The proposed contribution must be an identifiable history-weighted term beyond the best conventional model.

Information thermodynamics provides an equally strict constraint. Information is physical where it is instantiated; erasure has costs determined by the actual protocol, bath and correlations. A Verrell-specific thermal result would not be heat itself, but a preregistered, intervention-sensitive relationship between physical transition weight, retained history and later dissipation or selection.

Quantum sensors close part of the metrological gap, but no instrument replaces a physical prediction. NV centres, nanoSQUIDs and microcalorimeters become relevant only after the candidate source, scale, bandwidth and coupling path are stated. The next serious step is therefore a compact engineered-reservoir experiment with matched present states, different controlled histories, a fixed kernel model and an ablation that can make the effect disappear.

FINAL FORMULATION

History becomes physically active where retained information changes the accessible paths of a system. Verrell's Law predicts that the strength of this influence is not always uniform: sufficiently weighted transitions may leave a disproportionate, measurable contribution to later dynamics.

AUTHOR AND CITATION

Verrell's Law was originated by Marcos Verrell Moss Ross (M.R.). This information-physics edition develops the law as a proposed non-Markovian framework while retaining authorship solely with its originator.

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REFERENCES

1. Bérut, A., Arakelyan, A., Petrosyan, A., Ciliberto, S., Dillenschneider, R., & Lutz, E. (2012). Experimental verification of Landauer's principle linking information and thermodynamics. *Nature*, 483, 187-189. <https://doi.org/10.1038/nature10872>
2. Breuer, H.-P., Laine, E.-M., & Piilo, J. (2009). Measure for the degree of non-Markovian behavior of quantum processes in open systems. *Physical Review Letters*, 103, 210401. <https://doi.org/10.1103/PhysRevLett.103.210401>
3. Breuer, H.-P., Laine, E.-M., Piilo, J., & Vacchini, B. (2016). Colloquium: Non-Markovian dynamics in open quantum systems. *Reviews of Modern Physics*, 88, 021002. <https://doi.org/10.1103/RevModPhys.88.021002>
4. Dago, S., & Bellon, L. (2026). Shifting erasure cost below the Landauer bound with a demon-biased thermal bath. *Physical Review Letters*, accepted 8 September 2026. <https://doi.org/10.1103/rw6w-bskc>
5. Degen, C. L., Reinhard, F., & Cappellaro, P. (2017). Quantum sensing. *Reviews of Modern Physics*, 89, 035002. <https://doi.org/10.1103/RevModPhys.89.035002>
6. Ivander, F., Lindoy, L. P., & Lee, J. (2024). Unified framework for open quantum dynamics with memory. *Nature Communications*, 15, 8087. <https://doi.org/10.1038/s41467-024-52081-3>
7. Katsumi, R., Takada, K., Kawai, K., et al. (2025). High-sensitivity nanoscale quantum sensors based on a diamond micro-resonator. *Communications Materials*, 6, 49. <https://doi.org/10.1038/s43246-025-00770-x>
8. Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3), 183-191. <https://doi.org/10.1147/rd.53.0183>
9. Mori, H. (1965). Transport, collective motion, and Brownian motion. *Progress of Theoretical Physics*, 33(3), 423-455. <https://doi.org/10.1143/PTP.33.423>
10. Nakajima, S. (1958). On quantum theory of transport phenomena. *Progress of Theoretical Physics*, 20(6), 948-959. <https://doi.org/10.1143/PTP.20.948>
11. Piilo, J., Harter, A., Bergholm, V., et al. (2011). Experimental control of the transition from Markovian to non-Markovian dynamics of open quantum systems. *Nature Physics*, 7, 931-934. <https://doi.org/10.1038/nphys2085>
12. Pollock, F. A., Rodriguez-Rosario, C., Frauenheim, T., Paternostro, M., & Modi, K. (2018). Non-Markovian quantum processes: Complete framework and efficient characterization. *Physical Review A*, 97, 012127. <https://doi.org/10.1103/PhysRevA.97.012127>
13. Shimizu, T., Chida, K., Yamahata, G., & Nishiguchi, K. (2026). Thermodynamic constraints in dynamic random-access memory cells: Experimental verification of energy efficiency limits in information erasure. *Physical Review Letters*, 136, 117103. <https://doi.org/10.1103/1sgm-dhys>
14. Vasyukov, D., Anahory, Y., Embon, L., et al. (2013). A scanning superconducting quantum interference device with single electron spin sensitivity. *Nature Nanotechnology*, 8, 639-644. <https://doi.org/10.1038/nnano.2013.169>
15. Yan, L.-L., Xiong, T.-P., Rehan, K., et al. (2018). Single-atom demonstration of the quantum Landauer principle. *Physical Review Letters*, 120, 210601. <https://doi.org/10.1103/PhysRevLett.120.210601>
16. Zwanzig, R. (1961). Memory effects in irreversible thermodynamics. *Physical Review*, 124(4), 983-992. <https://doi.org/10.1103/PhysRev.124.983>

REFERENCE AND CLAIM NOTE

The references establish the surrounding physics and the present capabilities of relevant measurement platforms. None of the cited authors is represented as endorsing Verrell's Law. Established memory effects, quantum sensing and information-erasure experiments are scientific baselines; they become evidence for this framework only if a Verrell-specific, preregistered prediction is tested and survives its controls.